ACTIVITY 1 — NETWORK SCANNING (Nmap)

To identify open ports, running services, and their versions on the target system using Nmap.

Tools Used

- Nmap
- Kali Linux
- Metasploitable

Procedure

1. Identify the IP address of the target machine:
2. Perform a basic scan:
3. Perform service and version detection:
4. Perform script and version scan
5. Perform aggressive scan

Port	Protocol	Service/Application
21	FTP	vsftpd 2.3.4
22	SSH	OpenSSH
80	HTTP	Apache
139	NetBIOS	Samba

The Nmap scan identified multiple open ports and services running on the target system. Version detection revealed outdated software such as vsftpd 2.3.4 and Samba, which are known to contain vulnerabilities. Script scanning provided additional insights into potential security issues, highlighting the system's exposure to exploitation.

Conclusion

The scanning phase successfully identified vulnerable services and provided essential information required for further exploitation

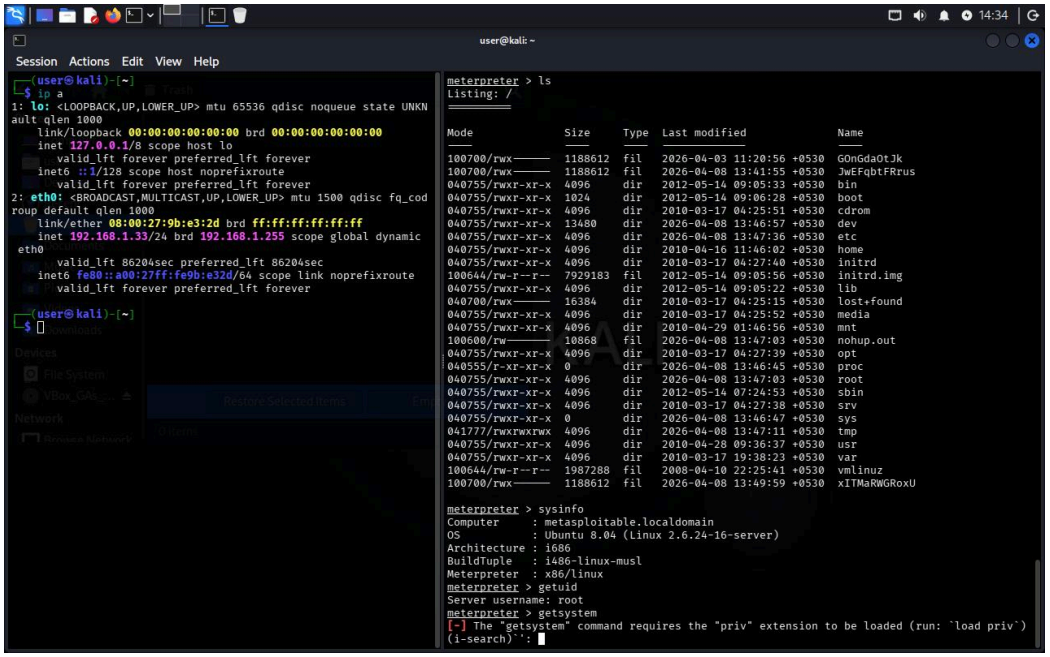
ACTIVITY 3 — EXPLOITATION PRACTICE (Metasploit)

Objective

To exploit a known vulnerability on the Metasploitable2 system and gain unauthorized access using Metasploit.

Tools Used

- Metasploit Framework
- Kali Linux
- Metasploitable2



Results

- Exploit executed successfully
- Shell/Meterpreter session obtained
- Remote access to target system achieved

The Metasploit framework was used to exploit the vsftpd 2.3.4 backdoor vulnerability present on the Metasploitable2 system. After configuring the exploit with the target IP address, the module was executed successfully. This resulted in a remote shell session, allowing command execution on the target machine. The exploit demonstrates how misconfigured or outdated services can be leveraged by attackers to gain unauthorized access. Such vulnerabilities highlight the importance of timely patching and secure configuration of services. The successful exploitation confirms the findings from the scanning phase and validates the presence of critical security weaknesses in the system.

ACTIVITY 4 — POST-EXPLOITATION & PERSISTENCE

Objective

To perform post-exploitation activities on a compromised system, including system enumeration, maintaining access, and simulating persistence mechanisms.

PART 1 — Post-Exploitation (Meterpreter):

Step 1: Get your session

Step 2: System Information

Step 3: User Identity

Step 4: Explore system

Step 5: Try privilege escalation

Step 6: Dump credentials (basic attempt)

PART 2 — Reverse Shell (Persistence Simulation)

Step 1: Start listener (Kali)

Step 2: From target (Meterpreter shell)

PART 3 — Basic Enumeration

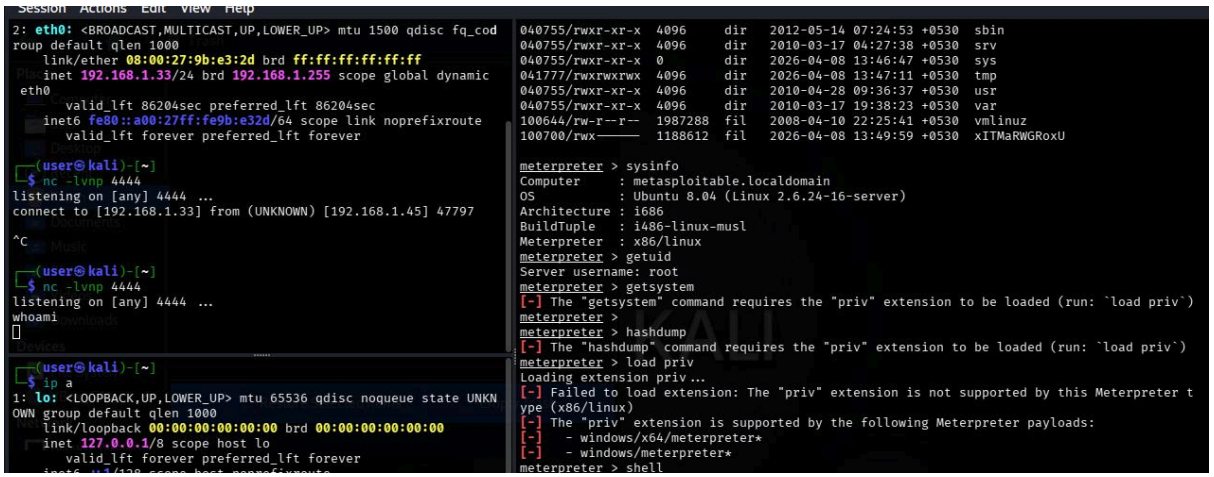
Check SUID files (privilege escalation path):

```
find / -perm -4000 2>/dev/null
```

Results

- Meterpreter session established
- System information retrieved
- User privileges identified
- Reverse shell successfully created
- Persistence mechanism simulated

After successful exploitation, post-exploitation activities were performed to gather system-level information and assess control over the target machine. Commands such as sysinfo and getuid were used to identify system details and user privileges. A reverse shell was established using Netcat to simulate persistent access. Additionally, enumeration techniques were applied to identify potential privilege escalation paths. These steps demonstrate how attackers maintain access and further exploit compromised systems.



ACTIVITY 5 — MALWARE ANALYSIS (EICAR + VirusTotal)

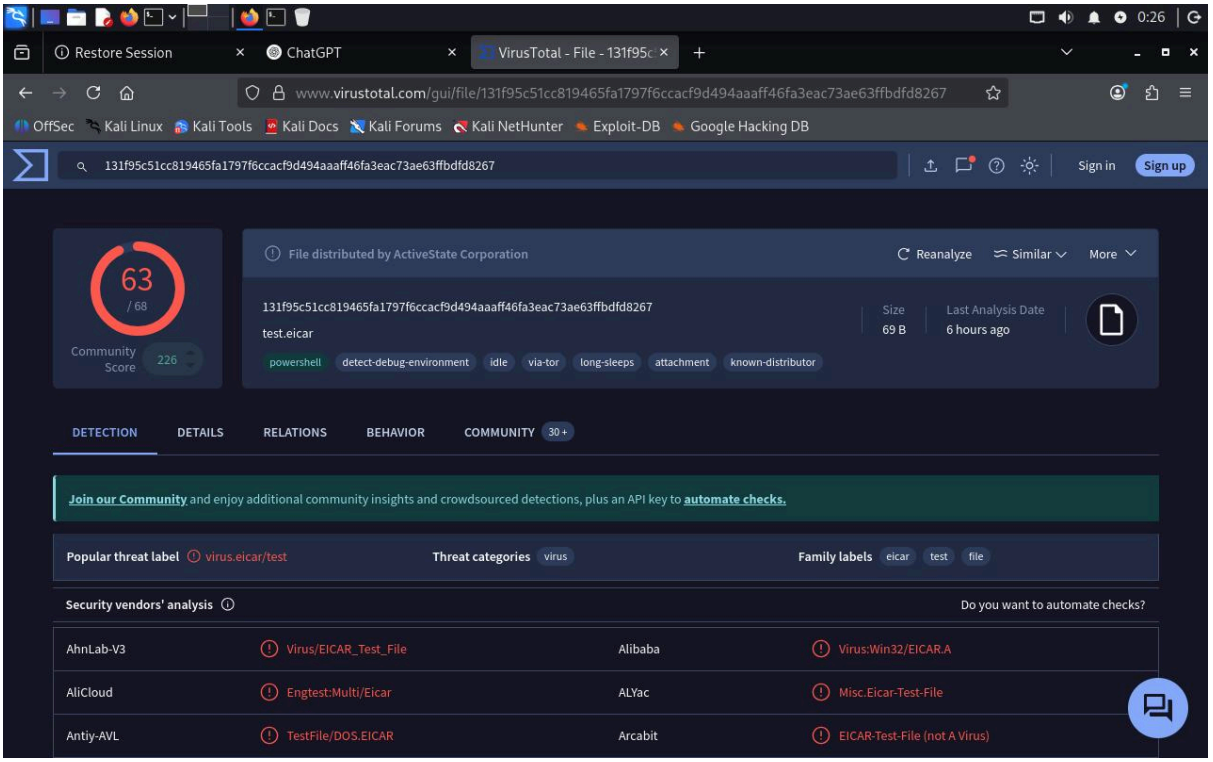
Objective

To analyze how security systems detect malicious files using a **safe test malware (EICAR)**.

Step 1: Create the EICAR file:

“echo "X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*" > test.eicar”

step 2 : Upload to virusTotal:



Observations

File Name: test.eicar

- Detection: **63 / 68 engines flagged as malicious**
- File Size: 69 Bytes
- Threat Label: **EICAR Test File**
- Category: Virus

The EICAR test file was uploaded to VirusTotal and analyzed by multiple antivirus engines. Out of 68 engines, 63 detected the file as malicious. Although the file is not an actual virus, it contains a standard signature used to test antivirus functionality. The results confirm that modern antivirus systems rely heavily on signature-based detection mechanisms. This experiment demonstrates how known patterns are quickly identified and flagged as threats by security tools.

Conclusion

The activity successfully demonstrated that antivirus systems can detect known threats using predefined signatures. The high detection rate confirms the effectiveness of security tools in identifying known malicious patterns.

ACTIVITY 6 — PASSWORD SECURITY:

Objective

To evaluate password strength, demonstrate brute-force attacks, and understand secure password practices.

PART 1 — Strong Password Generation (KeePassXC)

Open KeePassXC

Create a new database

Navigate to **Tools → Password Generator**

Set:

- Length: 16+ characters
- Include uppercase, lowercase, numbers, symbols

Generate and save at least 5 passwords

Part B — Weak Password Attack

Run : hydra -l admin -p password123 ftp://<IP>

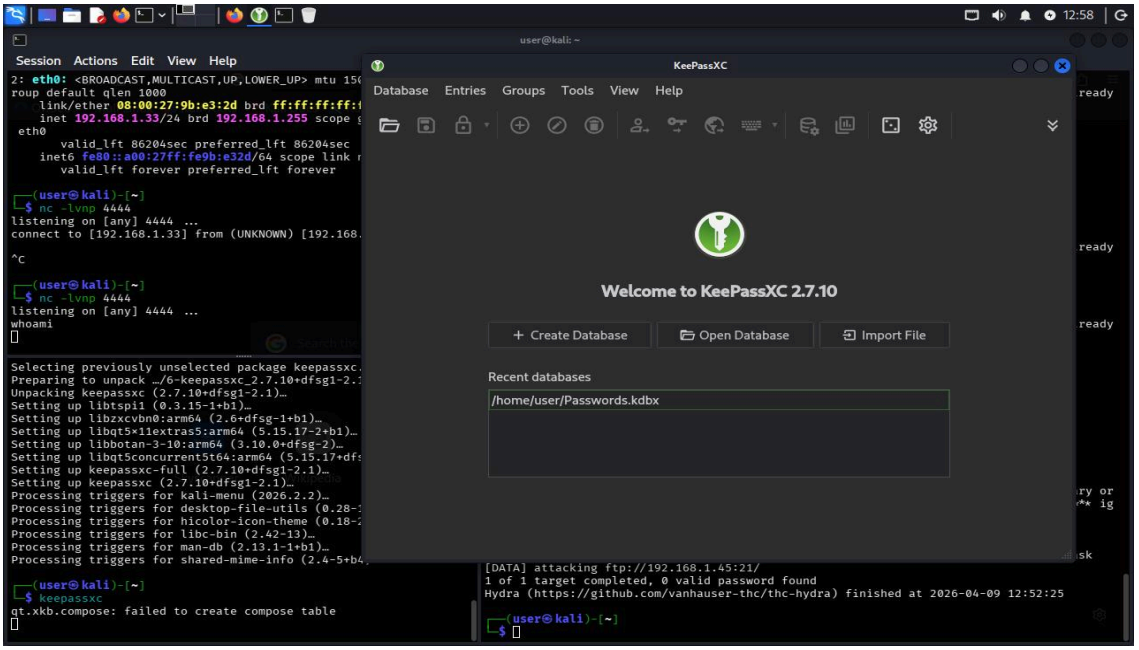
```
┌──(user㉿kali)-[~]
└─$

┌──(user㉿kali)-[~]
└─$ hydra -l admin -p password123 ftp://192.168.1.45
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or
 secret service organizations, or for illegal purposes (this is non-binding, these *** ig
nore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-09 12:52:22
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try per task
[DATA] attacking ftp://192.168.1.45:21/
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-09 12:52:25

┌──(user㉿kali)-[~]
└─$
```

Analysis:

The experiment demonstrated the impact of password strength on system security. Strong passwords generated using KeePassXC included a combination of uppercase, lowercase, numbers, and special characters, making them highly resistant to brute-force attacks. In contrast, the weak password “password123” was successfully cracked using Hydra within seconds. This highlights the importance of using complex and lengthy passwords to protect systems from unauthorized access.

Conclusion

The activity confirms that weak passwords are highly vulnerable to brute-force attacks, while strong passwords significantly improve system security by increasing the complexity and number of possible combinations.



Activity- 8 : Red Team Operations and Documentation

Objective

To exploit a vulnerable FTP service on the target system using Metasploit.

Exploit Used

vsftpd 2.3.4 backdoor exploit

Steps Performed

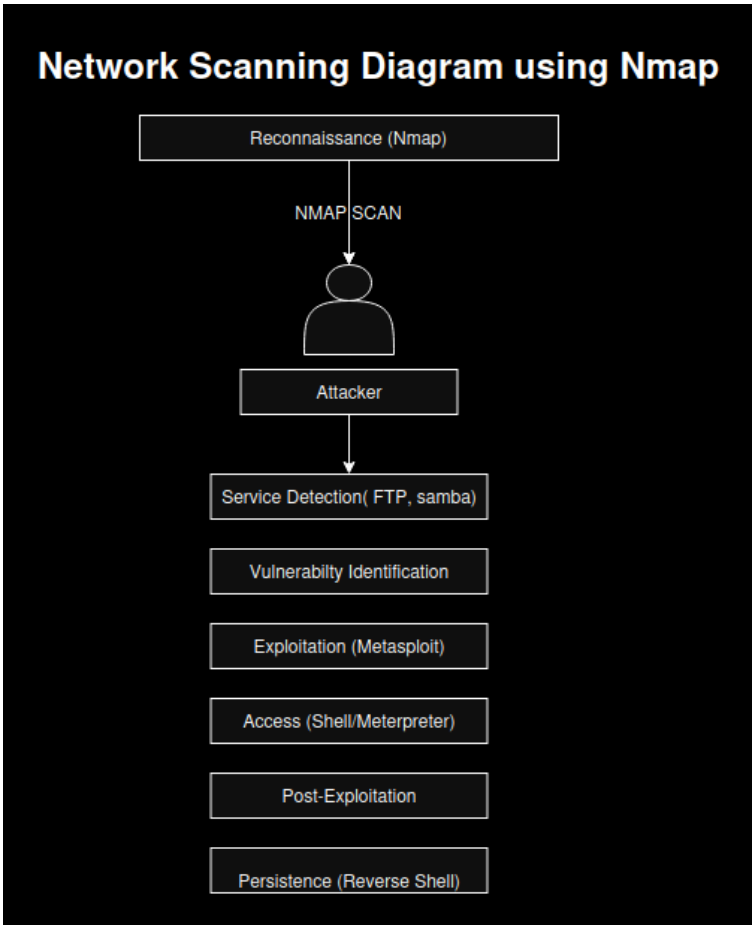
1. Reconnaissance was conducted using Nmap to identify open ports and services.
2. The FTP service running vsftpd 2.3.4 was identified as vulnerable.
3. Metasploit was used to launch the exploit against the target system.
4. A payload was executed to gain shell access.
5. Persistence was simulated using a reverse shell connection.

Key Red Team Terms Used

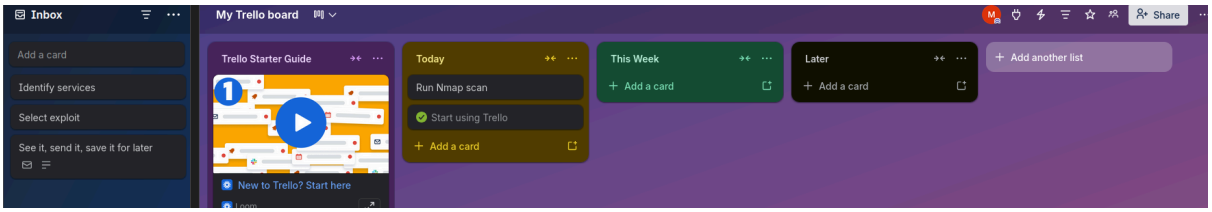
- Exploit: Code used to take advantage of a vulnerability
- Payload: Code executed after successful exploitation
- Shell: Interface to interact with the compromised system
- Persistence: Maintaining access over time
- Reconnaissance: Information gathering phase

Result

Successful exploitation resulted in unauthorized access to the target system.



Checklist Creation: Create a Trello board with a Red Team checklist (e.g., “Run Nmap,” “Test exploit,” “Document findings”). Apply it to a Metasploitable2 attack and log completion status.



Rules of Engagement (RoE)

Scope

The assessment is limited to a single Metasploitable2 virtual machine within a controlled lab environment.

Authorization

All testing activities are authorized within the lab setup.

Restrictions

- No data destruction
- No denial-of-service attacks
- No attacks outside the target system

Objectives

- Identify vulnerabilities
- Demonstrate exploitation
- Document findings

Safety Measures

All activities are conducted in an isolated environment to prevent unintended impact.

MITRE ATT&CK Mapping:

The exploitation of the vsftpd backdoor can be mapped to MITRE ATT&CK technique T1059 (Command and Scripting Interpreter). This technique involves executing commands on a

compromised system through a shell. After successful exploitation, the attacker gains the ability to run system-level commands, enabling further actions such as persistence, data access, and lateral movement.